Mathematical Properties of the Affine-DDPNM

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1 Preliminaries and notation

We adopt the domain partition, finite-element discretisation, and notation of the accompanying methodology document. The pore domain $\Omega \subset \mathbb{R}^3$ is partitioned into N non-overlapping subdomains $\{\Omega_i\}_{i=1}^N$ with m internal interfaces $\{\gamma_k\}_{k=1}^m$, assumed planar. Subdomain Ω_i has m_i ports, of which m_i^u are internal (unknown-traction) and m_i^k are boundary (known-traction). Each Ω_i is required to contain at least one positive-measure no-slip wall segment, ensuring that the local Stokes operator with mixed boundary conditions is invertible (no rigid-body modes).

Interface traction basis. On each interface γ, we introduce a local orthonormal frame $\{\boldsymbol{n}_\gamma, \boldsymbol{t}_{\gamma,1}, \boldsymbol{t}_{\gamma,2}\}$ with origin at the interface centroid $\boldsymbol{c}_\gamma$, and dimensionless in-plane coordinates

$$s(\boldsymbol{x}) = \frac{(\boldsymbol{x} - \boldsymbol{c}_\gamma) \cdot \boldsymbol{t}_{\gamma,1}}{a_\gamma}, \qquad t(\boldsymbol{x}) = \frac{(\boldsymbol{x} - \boldsymbol{c}_\gamma) \cdot \boldsymbol{t}_{\gamma,2}}{b_\gamma}, \tag{1}$$

where a_γ, b_γ are the half-spans of γ along the two tangent directions, ensuring $s, t = O(1)$ on γ under mild shape-regularity. The *mode index set* is

$$\mathcal{M} = \{0, s, t\} \times \{\boldsymbol{n}, \boldsymbol{t}_1, \boldsymbol{t}_2\}, \tag{2}$$

with scalar shape functions $\varphi_0 = 1$, $\varphi_s = s$, $\varphi_t = t$. For a subdomain Ω_i adjacent to γ, define the *signed* direction vectors

$$\boldsymbol{d}_{i,\gamma,\boldsymbol{n}} = \boldsymbol{n}_{i,\gamma}, \qquad \boldsymbol{d}_{i,\gamma,\boldsymbol{t}_\nu} = \sigma_{i,\gamma}\, \boldsymbol{t}_{\gamma,\nu} \ \ (\nu = 1, 2), \tag{3}$$

where $\sigma_{i,\gamma} = +1$ if Ω_i is the first member of the interface pair and $\sigma_{i,\gamma} = -1$ otherwise, satisfying $\boldsymbol{d}_{j,\gamma,\beta} = -\boldsymbol{d}_{i,\gamma,\beta}$ for the two pores i, j sharing γ. Throughout we abbreviate $\boldsymbol{d}_{i,\gamma,\beta}$.

The *pseudostress* $\boldsymbol{\sigma}(\boldsymbol{u}, p) := -p\mathbf{I} + \nu \nabla \boldsymbol{u}$ is used throughout. On each interface γ, the applied traction is expanded as

$$-\boldsymbol{\sigma}(\boldsymbol{u}, p)\, \boldsymbol{n}_{i,\gamma} = \sum_{(\alpha,\beta)\in\mathcal{M}} \lambda_{\gamma,\alpha,\beta}\, \varphi_\alpha(s,t)\, \boldsymbol{d}_{i,\gamma,\beta}, \tag{4}$$

where the minus sign on the left follows the classic convention. The nine coefficients $\boldsymbol{\lambda}_\gamma = (\lambda_{\gamma,0,\boldsymbol{n}}, \ldots, \lambda_{\gamma,t,\boldsymbol{t}_2}) \in \mathbb{R}^9$ are the unknown interface degrees of freedom.

For a subdomain Ω_i, let $n_i = 9m_i^u + m_i^k$ be the total number of local traction unknowns and $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$ the vector of all local coefficients. The Boolean restriction maps $R_i^u \in \{0,1\}^{n_i^u \times 9m}$ and $R_i^k \in \{0,1\}^{m_i^k \times m^k}$ and the global unknown $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$ are defined with block size 9×9 per interface.

Unit-response loads. For each mode $(\alpha, \beta) \in \mathcal{M}$ on an internal interface $\gamma_{i,r}$ of Ω_i, the Neumann load vector $\boldsymbol{f}_{i,r}^{(\alpha,\beta)} \in \mathbb{R}^{n_{i,u}}$ has entries

$$\left[\boldsymbol{f}_{i,r}^{(\alpha,\beta)}\right]_j = \int_{\gamma_{i,r}} \varphi_\alpha(s,t)\, \boldsymbol{d}_{i,\gamma,\beta} \cdot \boldsymbol{\phi}_j^{(i)}\, \mathrm{d}S, \tag{5}$$

where $\{\boldsymbol{\phi}_j^{(i)}\}$ is the local velocity basis. By the sign convention (4), a positive coefficient λ produces a traction directed *into* the pore; the load vector carries no extra minus sign—it is the right-hand side of the discrete momentum equation. For boundary ports, only the mode $(\alpha, \beta) = (0, \boldsymbol{n})$ is used with a *prescribed* coefficient equal to the inlet/outlet pressure.

Local Stokes system. For each unit load, the discrete Taylor–Hood $[P_2]^3$–P_1 system reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & -\varepsilon M_{p,i} \end{bmatrix} \begin{bmatrix} \mathbf{U}_{i,r}^{(\alpha,\beta)} \\ \mathbf{X}_{i,r}^{(\alpha,\beta)} \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_{i,r}^{(\alpha,\beta)} \\ \mathbf{0} \end{bmatrix}, \tag{6}$$

where A_i is SPD, B_i is the divergence matrix, $\varepsilon \geq 0$ is an optional pressure stabilisation parameter, and $M_{p,i}$ is the pressure mass matrix. In what follows we set $\varepsilon = 0$ (no stabilisation), so that exact discrete incompressibility $B_i \mathbf{U}_i = \mathbf{0}$ holds. All numerical results reported use $\varepsilon = 0$; for $\varepsilon > 0$ the mass conservation statement in Theorem 2.6 requires a modification.

Local compliance map. The local Neumann-to-Dirichlet (compliance) map $\mathbf{G}_i \in \mathbb{R}^{n_i \times n_i}$ has entries

$$\mathbf{G}_i\big[(\gamma, \alpha, \beta),\, (\gamma', \alpha', \beta')\big] = \big[\boldsymbol{f}_i^{(\alpha,\beta)}(\gamma)\big]^\top \mathbf{U}_i^{(\alpha',\beta')}(\gamma'), \tag{7}$$

where the index triple (γ, α, β) ranges over the n_i modes. $\mathbf{G}_i$ is symmetric positive semi-definite (proved in Lemma 2.3). It is partitioned as

$$\mathbf{G}_i = \begin{bmatrix} \mathbf{G}_i^{uu} & \mathbf{G}_i^{uk} \\ \mathbf{G}_i^{ku} & \mathbf{G}_i^{kk} \end{bmatrix},$$

where $\mathbf{G}_i^{uu} \in \mathbb{R}^{9m_i^u \times 9m_i^u}$ maps unknown tractions, $\mathbf{G}_i^{uk} \in \mathbb{R}^{9m_i^u \times m_i^k}$ couples unknown to known, and $\mathbf{G}_i^{kk} \in \mathbb{R}^{m_i^k \times m_i^k}$ involves boundary ports only.

Weighted velocity moments. For each mode (α, β) on interface $\gamma_{i,r}$, define

$$M_{i,r}^{(\alpha,\beta)} = \int_{\gamma_{i,r}} \varphi_\alpha(s,t)\, \boldsymbol{u}_{i,h} \cdot \boldsymbol{d}_{i,\gamma,\beta}\, \mathrm{d}S. \tag{8}$$

Only $M_{i,r}^{(0,\boldsymbol{n})} = \int_\gamma \boldsymbol{u}{\cdot}\boldsymbol{n}\, \mathrm{d}S$ is the volumetric flow rate. The remaining eight quantities are weighted velocity moments (linear normal flux moments $M^{(s,\boldsymbol{n})}, M^{(t,\boldsymbol{n})}$ and tangential velocity moments $M^{(\alpha,\boldsymbol{t}_\nu)}$). From $\boldsymbol{d}_{j,\gamma,\beta} = -\boldsymbol{d}_{i,\gamma,\beta}$ we obtain the consistency condition $M_{i,r}^{(\alpha,\beta)} + M_{j,r}^{(\alpha,\beta)} = 0$ for any continuous velocity field across γ_r.

Global Schur complement. Enforcing continuity of all nine weighted velocity moments on every internal interface yields the global system

$$\mathbf{S}\,\boldsymbol{\lambda} = \boldsymbol{F}, \qquad \mathbf{S} = \sum_{i=1}^{N} (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \qquad \boldsymbol{F} = -\sum_{i=1}^{N} (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k, \tag{9}$$

where $\boldsymbol{\lambda}^k \in \mathbb{R}^{m^k}$ collects prescribed boundary tractions. $\mathbf{S} \in \mathbb{R}^{9m \times 9m}$ is symmetric; its positive definiteness is established in Theorem 2.6 under an anchoring condition.

2 Mathematical properties

Lemma 2.1 (Local well-posedness). *Assume each Ω_i contains a positive-measure no-slip wall segment. Then for every interface-traction vector $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$, the discrete Stokes subproblem (6) in Ω_i with $\varepsilon = 0$ admits a unique solution $(\mathbf{U}_i, \mathbf{X}_i)$.*

Proof. The matrix A_i is SPD. With the wall Dirichlet condition, the Taylor–Hood pair satisfies the discrete inf–sup condition on Ω_i, so B_i has full row rank and the Schur complement $B_i A_i^{-1} B_i^\top$ is SPD. Hence the saddle-point matrix K_i is invertible. The right-hand side of (6) is a linear combination of the primitive load vectors; a unique solution exists for any $\boldsymbol{\lambda}_i$. □

Lemma 2.2 (Discrete incompressibility of unit responses)**.** *For each mode $(\alpha, \beta) \in \mathcal{M}$ on an internal interface $\gamma_{i,r}$, the response velocity satisfies $B_i\, \mathbf{U}_{i,r}^{(\alpha,\beta)} = \mathbf{0}$.*

Proof. Immediate from the second block row of (6) with $\varepsilon = 0$. For reference we also give the explicit form: $\mathbf{U}_{i,r}^{(\alpha,\beta)} = H_i \boldsymbol{f}_{i,r}^{(\alpha,\beta)}$ where

$$H_i := A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}, \qquad M_i := B_i A_i^{-1} B_i^\top, \tag{10}$$

and $B_i H_i = 0$ is verified directly. □

Lemma 2.3 (Symmetry and energy identity)**.** *$\mathbf{G}_i$ defined by (7) is symmetric positive semi-definite. For any $\boldsymbol{\lambda}_i \in \mathbb{R}^{n_i}$,*

$$\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = \mathbf{U}_i^\top A_i \mathbf{U}_i \geq 0, \tag{11}$$

where $\mathbf{U}_i = \sum_{\gamma,\alpha,\beta} \lambda_{i,\gamma,\alpha,\beta}\, \mathbf{U}_i^{(\alpha,\beta)}(\gamma)$ is the velocity field induced by $\boldsymbol{\lambda}_i$.

Proof. Let $\mu = (\gamma, \alpha, \beta)$, $\mu' = (\gamma', \alpha', \beta')$. From Lemma 2.2 and (10), $\mathbf{U}_i^{(\mu)} = H_i \boldsymbol{f}_i^{(\mu)}$. The identity $H_i^\top A_i H_i = H_i$ follows from direct expansion:

$$\begin{aligned} H_i^\top A_i H_i &= (A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1})\, A_i\, (A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}) \\ &= A_i^{-1} - 2A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1} + A_i^{-1} B_i^\top M_i^{-1} (B_i A_i^{-1} B_i^\top) M_i^{-1} B_i A_i^{-1} \\ &= A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1} = H_i, \end{aligned}$$

since $B_i A_i^{-1} B_i^\top = M_i$. Now:

$$\begin{aligned} \mathbf{G}_i(\mu, \mu') &= \big[\boldsymbol{f}_i^{(\mu)}\big]^\top \mathbf{U}_i^{(\mu')} = \big[\boldsymbol{f}_i^{(\mu)}\big]^\top H_i \boldsymbol{f}_i^{(\mu')} \\ &= \big[\boldsymbol{f}_i^{(\mu)}\big]^\top (H_i^\top A_i H_i) \boldsymbol{f}_i^{(\mu')} = \big[\mathbf{U}_i^{(\mu)}\big]^\top A_i \mathbf{U}_i^{(\mu')}. \end{aligned}$$

Symmetry follows from symmetry of A_i. Summing over μ, μ' with coefficients $\lambda_{i,\mu}$ yields $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = \big(\sum_\mu \lambda_{i,\mu} \mathbf{U}_i^{(\mu)}\big)^\top A_i \big(\sum_{\mu'} \lambda_{i,\mu'} \mathbf{U}_i^{(\mu')}\big) = \mathbf{U}_i^\top A_i \mathbf{U}_i \geq 0$. □

Lemma 2.4 (Kernel of the local compliance map)**.** *Let $\mathbf{1}_{(0,\boldsymbol{n})} \in \mathbb{R}^{n_i}$ be the vector with entry 1 at every uniform-normal mode $(\gamma, 0, \boldsymbol{n})$ (internal interfaces and boundary ports alike) and entry 0 elsewhere. Let $L_i \in \mathbb{R}^{n_{i,u} \times n_i}$ be the matrix whose columns are all primitive load vectors $\boldsymbol{f}_i^{(\mu)}$, so that $\mathbf{G}_i = L_i^\top H_i L_i$. If*

$$\operatorname{rank}(H_i L_i) = n_i - 1, \tag{12}$$

then $\ker(\mathbf{G}_i) = \operatorname{span}\{\mathbf{1}_{(0,\boldsymbol{n})}\}$ is exactly one-dimensional.

Proof. **(i) $\mathbf{1}_{(0,\boldsymbol{n})} \in \ker(\mathbf{G}_i)$.** Set all uniform-normal coefficients to p_{const} and all others to zero. Summing the loads:

$$\begin{aligned} \sum_\gamma \boldsymbol{f}_i^{(\gamma,0,\boldsymbol{n})} &= \sum_\gamma \int_{\gamma_{i,\gamma}} \boldsymbol{\phi}_j^{(i)} \cdot \boldsymbol{n}_{i,\gamma}\, \mathrm{d}S = \int_{\partial\Omega_i} \boldsymbol{\phi}_j^{(i)} \cdot \boldsymbol{n}\, \mathrm{d}S \\ &= \int_{\Omega_i} \nabla \cdot \boldsymbol{\phi}_j^{(i)}\, \mathrm{d}\boldsymbol{x} = \big[B_i^\top \mathbf{1}\big]_j, \end{aligned}$$

where $\mathbf{1}$ is the coefficient vector of $p_h \equiv 1$. Hence $(\mathbf{U}_i, \mathbf{X}_i) = (\mathbf{0}, p_{\text{const}}\mathbf{1})$ solves (6), and by uniqueness $\mathbf{U}_i = \mathbf{0}$. The energy identity gives $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = 0$, so $\mathbf{G}_i \boldsymbol{\lambda}_i = \mathbf{0}$ by positive semi-definiteness.

(ii) $\ker(\mathbf{G}_i) \subseteq \operatorname{span}\{\mathbf{1}_{(0,\boldsymbol{n})}\}$**.** Since $\mathbf{G}_i = L_i^\top H_i L_i$ with H_i symmetric, $\ker(\mathbf{G}_i) = \ker(H_i L_i)$ because H_i is SPD on $\operatorname{Range}(L_i)$. From (12), $\dim\ker(H_i L_i) = n_i - \operatorname{rank}(H_i L_i) = 1$. Part (i) shows $\mathbf{1}_{(0,\boldsymbol{n})} \in \ker(H_i L_i)$, hence $\ker(\mathbf{G}_i) = \operatorname{span}\{\mathbf{1}_{(0,\boldsymbol{n})}\}$. □ □

Remark 2.5. *Assumption (12) states that the n_i primitive responses $H_i \boldsymbol{f}_i^{(\mu)}$ span a subspace of codimension 1 in $\operatorname{Range}(H_i)$, the sole dependency being the constant-pressure relation $\sum_\gamma \boldsymbol{f}_i^{(\gamma,0,\boldsymbol{n})} = B_i^\top \mathbf{1}$. This is verified numerically for all geometries tested (the local compliance matrices have exactly one zero eigenvalue to machine precision) and holds analytically provided the nine modes per interface are linearly independent on each face and no accidental linear dependencies arise from the load assembly. A rigorous verification for general Lipschitz pore geometries requires further work; the numerical evidence is uniform across all tested configurations.*

Theorem 2.6 (Solvability and mass conservation of Affine-DDPNM)**.** *Let $\mathbf{S} \in \mathbb{R}^{9m\times 9m}$ be defined by (9) and assume Lemma 2.4 holds for every pore. Further assume that every connected component of the pore adjacency graph contains at least one pore with a prescribed-traction (inlet/outlet) boundary port (the* anchoring condition*). Then:*

(i) $\mathbf{S}$ *is symmetric positive-definite (SPD).*

(ii) The system $\mathbf{S}\boldsymbol{\lambda} = \boldsymbol{F}$ has a unique solution.

(iii) With $\varepsilon = 0$, the reconstructed velocity field satisfies local and global discrete mass conservation $\sum_r M_{i,r}^{(0,\boldsymbol{n})} = 0$ (each pore) and $\int_{\partial\Omega} \boldsymbol{u}_h^{DD} \cdot \boldsymbol{n}\,\mathrm{d}S = 0$.

Proof. **(i) SPD.** Symmetry follows from Lemma 2.3. For any $\boldsymbol{\lambda} \in \mathbb{R}^{9m}$, let $\boldsymbol{\lambda}_i^u = R_i^u \boldsymbol{\lambda}$. By (11), $\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = \sum_i (\boldsymbol{\lambda}_i^u)^\top \mathbf{G}_i^{uu} \boldsymbol{\lambda}_i^u = \sum_i \mathbf{U}_i^\top A_i \mathbf{U}_i \geq 0$.

If $\boldsymbol{\lambda}^\top \mathbf{S} \boldsymbol{\lambda} = 0$, then $\mathbf{U}_i = \mathbf{0}$ for every i. By Lemma 2.4, each $\boldsymbol{\lambda}_i^u$ equals $c_i \mathbf{1}_{(0,\boldsymbol{n})}$ for some $c_i \in \mathbb{R}$, restricted to the internal interfaces of Ω_i. Continuity of $\boldsymbol{\lambda}$ across interfaces forces $c_i = c$ for all pores in the same connected component of the pore graph. For a component containing a prescribed-traction port, the boundary entry of $\mathbf{1}_{(0,\boldsymbol{n})}$ is *not* included in $\boldsymbol{\lambda}$ (boundary ports belong to $\boldsymbol{\lambda}^k$, not $\boldsymbol{\lambda}$); hence $c = 0$ on that component. The anchoring condition ensures every component contains a boundary port, so $c = 0$ globally. Hence $\boldsymbol{\lambda} = \mathbf{0}$, proving $\mathbf{S}$ is SPD.

(ii) Immediate from (i).

(iii) Mass conservation. With $\varepsilon = 0$, the discrete incompressibility $B_i \mathbf{U}_i = \mathbf{0}$ holds. Choosing the constant test function $q_h \equiv 1 \in Q_{i,h}$ and applying the divergence theorem yields $\sum_r M_{i,r}^{(0,\boldsymbol{n})} = 0$. The Schur assembly enforces $M_{i,r}^{(0,\boldsymbol{n})} + M_{j,r}^{(0,\boldsymbol{n})} = 0$ on each internal interface; summing over all pores, internal fluxes cancel and only boundary terms remain, giving the global balance. □

Remark 2.7 (Anchoring in practice)**.** *The anchoring condition is satisfied for all benchmark geometries studied: the uniform and random sphere packings have inlet/outlet boundaries cutting through the entire domain, and every pore in the watershed partition that lacks a solid-wall facet is merged into its neighbour (the* floating-basin merge *procedure). This merge step is not an implementation detail—it is the practical enforcement of the anchoring condition and is required for the Schur system to be non-singular.*

A Error estimate as a Galerkin projection

We now characterise the Affine-DDPNM as a Galerkin restriction of a broken-pressure non-overlapping domain-decomposition (C-DD) formulation. The reference C-DD system uses *discontinuous* pressure across subdomains ($Q_h^{\mathrm{br}} = \bigoplus_i Q_{i,h}$), not the global continuous Taylor–Hood space. Consequently, the error estimate below quantifies the distance from this broken-pressure C-DD solution, not from the monolithic Taylor–Hood FEM. Numerically the two references differ by $O(10^{-12})$ on the tested meshes (the exact FE-trace Schur oracle, a primal mixed Schur complement on the global continuous space, coincides with the global solve to machine precision), but the Galerkin projection proof itself only covers the broken-pressure C-DD.

A.1 C-DD formulation (broken pressure)

On each internal interface $\gamma_{i,r}$, the traction is expanded in the FE face basis $\{\psi_\ell\}$ (the restriction of the velocity trace basis):

$$\boldsymbol{t}_{i,r}^{\mathrm{FE}}(\boldsymbol{x}) = \sum_{\ell=1}^{m_{i,r}} \left(\eta_{i,r}^{\ell} \boldsymbol{n}_{i,r} + \tau_{i,r}^{\ell,1} \boldsymbol{t}_{i,r}^{1} + \tau_{i,r}^{\ell,2} \boldsymbol{t}_{i,r}^{2}\right) \psi_\ell(\boldsymbol{x}),$$

with $m_{i,r}$ velocity DOFs on $\gamma_{i,r}$. Collect the nodal values into $\boldsymbol{\eta}_{i,r}, \boldsymbol{\tau}_{i,r}^1, \boldsymbol{\tau}_{i,r}^2 \in \mathbb{R}^{m_{i,r}}$, and the full C-DD traction on $\gamma_{i,r}$ into $\boldsymbol{\xi}_{i,r} := (\boldsymbol{\eta}_{i,r}; \boldsymbol{\tau}_{i,r}^1; \boldsymbol{\tau}_{i,r}^2) \in \mathbb{R}^{3m_{i,r}}$.

The Neumann coupling blocks are

$$\begin{aligned} N_{i,r}^{\boldsymbol{n}} &\in \mathbb{R}^{m_{i,r} \times n_{i,u}}, & [N_{i,r}^{\boldsymbol{n}}]_{\ell,j} &= \int_{\gamma_{i,r}} \psi_\ell \, (\boldsymbol{\phi}_j^{(i)} \cdot \boldsymbol{n}_{i,r}) \, \mathrm{d}S, \\ N_{i,r}^{\boldsymbol{t}_\nu} &\in \mathbb{R}^{m_{i,r} \times n_{i,u}}, & [N_{i,r}^{\boldsymbol{t}_\nu}]_{\ell,j} &= \int_{\gamma_{i,r}} \psi_\ell \, (\boldsymbol{\phi}_j^{(i)} \cdot \boldsymbol{t}_{i,r}^{\nu}) \, \mathrm{d}S \quad (\nu = 1, 2). \end{aligned}$$

Stacking all interfaces vertically gives the local C-DD coupling matrix

$$N_i = \begin{bmatrix} N_i^{\boldsymbol{n}} \\ N_i^{\boldsymbol{t}_1} \\ N_i^{\boldsymbol{t}_2} \end{bmatrix} \in \mathbb{R}^{3\sum_r m_{i,r} \times n_{i,u}}, \tag{13}$$

and the local traction $\boldsymbol{\xi}_i = (\boldsymbol{\eta}_i; \boldsymbol{\tau}_i^1; \boldsymbol{\tau}_i^2)$.

The discrete Stokes system on Ω_i reads

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} V_i \\ P_i \end{bmatrix} = - \begin{bmatrix} N_i^\top \boldsymbol{\xi}_i \\ \mathbf{0} \end{bmatrix} + (\text{boundary contributions}).$$

Eliminating interior DOFs yields the local C-DD Schur complement $\widehat{\mathbf{S}}_i = N_i H_i N_i^\top$. Global Boolean assembly with maps $\widehat{R}_i^u$ gives

$$\widehat{\mathbf{S}} \, \boldsymbol{\xi}^{\mathrm{FE}} = \widehat{\boldsymbol{F}}, \qquad \widehat{\mathbf{S}} = \sum_{i=1}^{N} (\widehat{R}_i^u)^\top \widehat{\mathbf{S}}_i \widehat{R}_i^u, \tag{14}$$

with $\boldsymbol{\xi}^{\mathrm{FE}} \in \mathbb{R}^{m_\Gamma}$ collecting all global interface nodal tractions.

A.2 Affine restriction and equivalence

For each interface γ_k, let $M_k \in \mathbb{R}^{m_k \times m_k}$ be the face mass matrix. For each mode $(\alpha, \beta) \in \mathcal{M}$, define $\boldsymbol{b}_k^{\alpha,\beta} \in \mathbb{R}^{m_k}$ by $(\boldsymbol{b}_k^{\alpha,\beta})_\ell = \int_{\gamma_k} \varphi_\alpha(s,t) \, \psi_\ell \, \mathrm{d}S$. The per-mode L^2-projection vector is $P_k^{\alpha,\beta} = M_k^{-1} \boldsymbol{b}_k^{\alpha,\beta} \in \mathbb{R}^{m_k}$.

The local restriction matrix $P_k^{\mathrm{aff}} \in \mathbb{R}^{3m_k \times 9}$ maps the nine affine coefficients to the three-component nodal traction:

$$P_k^{\mathrm{aff}} = \begin{bmatrix} P_k^{0,\boldsymbol{n}} & P_k^{0,\boldsymbol{t}_1} & P_k^{0,\boldsymbol{t}_2} & P_k^{s,\boldsymbol{n}} & P_k^{s,\boldsymbol{t}_1} & P_k^{s,\boldsymbol{t}_2} & P_k^{t,\boldsymbol{n}} & P_k^{t,\boldsymbol{t}_1} & P_k^{t,\boldsymbol{t}_2} \\ \mathbf{0} & P_k^{0,\boldsymbol{t}_1} & \mathbf{0} & \mathbf{0} & P_k^{s,\boldsymbol{t}_1} & \mathbf{0} & \mathbf{0} & P_k^{t,\boldsymbol{t}_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & P_k^{0,\boldsymbol{t}_2} & \mathbf{0} & \mathbf{0} & P_k^{s,\boldsymbol{t}_2} & \mathbf{0} & \mathbf{0} & P_k^{t,\boldsymbol{t}_2} \end{bmatrix}. \tag{15}$$

Each column occupies only the component block corresponding to its direction β:

- $\beta = \boldsymbol{n}$: only the first (normal) block-row is non-zero;
- $\beta = \boldsymbol{t}_1$: only the second block-row is non-zero;
- $\beta = \boldsymbol{t}_2$: only the third block-row is non-zero.

The global restriction operator is the block-diagonal assembly $\mathcal{P}_{\text{aff}} = \bigoplus_{k=1}^{m} P_k^{\text{aff}} : \mathbb{R}^{9m} \to \mathbb{R}^{m_\Gamma}$.

Lemma A.1 (Equivalence)**.** *Let* $\mathbf{S}$ *be the assembled Schur complement* (9) *and* $\mathcal{P}_{aff}$ *be defined as above. Then, for the internal degrees of freedom,*

$$\mathbf{S} = \mathcal{P}_{aff}^\top \widehat{\mathbf{S}}\, \mathcal{P}_{aff}, \qquad \boldsymbol{F} = \mathcal{P}_{aff}^\top \widehat{\boldsymbol{F}}, \tag{16}$$

where $\widehat{\mathbf{S}}$ *and* $\widehat{\boldsymbol{F}}$ *are the C-DD system* (14)*.*

Proof. Fix a pore Ω_i. For each internal interface $\gamma_{i,r}$ and each mode (α, β), the primitive load vector (5) equals (up to the sign absorbed in $\boldsymbol{d}_{i,\gamma,\beta}$)

$$\boldsymbol{f}_i^{(\gamma,\alpha,\beta)} = -(N_{i,r}^{\beta})^\top P_{i,r}^{\alpha,\beta},$$

where $N_{i,r}^{\beta}$ is the coupling block for direction β. Note that only the β-component of $P_{i,r}^{\text{aff}}$ is non-zero, so the sum over all three directions in the original erroneous formula reduces to the single term above. The matrix of all $9m_i^u$ primitive internal-interface loads is

$$L_i^u = -(N_i^u)^\top P_i^{\text{aff}},$$

where N_i^u is the C-DD coupling matrix restricted to internal interfaces and P_i^{aff} is the block-diagonal assembly of $\{P_{i,r}^{\text{aff}}\}_{r=1}^{m_i^u}$. The local compliance block is $\mathbf{G}_i^{uu} = (L_i^u)^\top H_i L_i^u = (P_i^{\text{aff}})^\top N_i^u H_i (N_i^u)^\top P_i^{\text{aff}} = (P_i^{\text{aff}})^\top \widehat{\mathbf{S}}_i^{uu} P_i^{\text{aff}}$.

Global Boolean assembly with R_i^u (mapping local internal modes to $\mathbb{R}^{9m}$) and the C-DD Boolean maps $\widehat{R}_i^u$ (mapping local C-DD nodal DOFs to $\mathbb{R}^{m_\Gamma}$) respects the commutative relation $\widehat{R}_i^u P_i^{\text{aff}} = \mathcal{P}_{\text{aff}}|_{\Omega_i} R_i^u$ (both sides map the same global affine DOFs to the same global C-DD DOFs). Summing over all pores yields (16). The boundary forcing term is treated analogously with prescribed $\boldsymbol{\lambda}^k$ replacing $\boldsymbol{\lambda}$. □

A.3 Error estimate

Theorem A.2 (Error estimate for Affine-DDPNM)**.** *Let* $\boldsymbol{\xi}^{FE}$ *solve the C-DD system* (14)*. Let* $\boldsymbol{\lambda}^{aff}$ *solve* $\mathbf{S}\,\boldsymbol{\lambda}^{aff} = \boldsymbol{F}$ *and set* $\boldsymbol{\xi}^{aff} = \mathcal{P}_{aff}\boldsymbol{\lambda}^{aff}$*. Then:*

(i) $\|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{aff}\|_{\widehat{\mathbf{S}}} = \min_{\boldsymbol{\mu}\in\mathbb{R}^{9m}} \|\boldsymbol{\xi}^{FE} - \mathcal{P}_{aff}\,\boldsymbol{\mu}\|_{\widehat{\mathbf{S}}}$*;*

(ii) $\sum_i \|\mathbf{U}_i^{FE} - \mathbf{U}_i^{aff}\|_{A_i}^2 = \|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{aff}\|_{\widehat{\mathbf{S}}}^2$*.*

Proof. By Lemma A.1, the Affine-DDPNM system is the Galerkin restriction of the C-DD system onto $\text{Range}(\mathcal{P}_{\text{aff}})$. Galerkin orthogonality gives $\mathcal{P}_{\text{aff}}^\top \widehat{\mathbf{S}}(\boldsymbol{\xi}^{\text{aff}} - \boldsymbol{\xi}^{\text{FE}}) = \mathbf{0}$. Hence $\boldsymbol{e} := \boldsymbol{\xi}^{\text{FE}} - \boldsymbol{\xi}^{\text{aff}}$ is $\widehat{\mathbf{S}}$-orthogonal to $\text{Range}(\mathcal{P}_{\text{aff}})$. For any $\boldsymbol{\mu} \in \mathbb{R}^{9m}$, the Pythagorean identity in the $\widehat{\mathbf{S}}$-inner product yields

$$\|\boldsymbol{\xi}^{\text{FE}} - \mathcal{P}_{\text{aff}}\,\boldsymbol{\mu}\|_{\widehat{\mathbf{S}}}^2 = \|\boldsymbol{e}\|_{\widehat{\mathbf{S}}}^2 + \|\boldsymbol{\xi}^{\text{aff}} - \mathcal{P}_{\text{aff}}\,\boldsymbol{\mu}\|_{\widehat{\mathbf{S}}}^2,$$

with equality at $\boldsymbol{\mu} = \boldsymbol{\lambda}^{\text{aff}}$ (the choice $\boldsymbol{\mu} = \boldsymbol{\lambda}^{\text{aff}}$ makes the last term zero by definition of $\boldsymbol{\xi}^{\text{aff}}$). This proves (i). Part (ii) follows from the C-DD energy identity $\|\boldsymbol{\xi} - \tilde{\boldsymbol{\xi}}\|_{\widehat{\mathbf{S}}}^2 = \sum_i \|\mathbf{U}_i(\boldsymbol{\xi}) - \mathbf{U}_i(\tilde{\boldsymbol{\xi}})\|_{A_i}^2$. □

A.4 Discussion

Remark A.3 (Enrichment and energy norm)**.** *The classic DD-PNM restriction* $\mathcal{P}_{P0} : \mathbb{R}^m \to \mathbb{R}^{m_\Gamma}$ L^2*-projects the constant normal function onto the FE face basis (normal component only). Because* $\text{Range}(\mathcal{P}_{P0}) \subset \text{Range}(\mathcal{P}_{aff})$*, Theorem A.2 guarantees* $\|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{aff}\|_{\widehat{\mathbf{S}}} \leq \|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{P0}\|_{\widehat{\mathbf{S}}}$*: the Affine error in the C-DD energy norm never exceeds the classic error. This monotonicity* *holds for the $\widehat{\mathbf{S}}$-energy norm; it does* not *automatically extend to the velocity L^2, pressure L^2, broken H^1, or outlet flux errors—those depend additionally on the specific geometry and on how the energy norm relates to L^2-type norms. The 73–90% L^2 reductions observed numerically are* consistent *with the energy-norm enrichment guarantee, but are not formally implied by it alone.*

Remark A.4 (Mesh dependence)**.** *The optimality result ((i)) bounds the discrete error $\|\boldsymbol{\xi}^{FE} - \boldsymbol{\xi}^{aff}\|_{\widehat{\mathbf{S}}_h}$ in the mesh-dependent energy norm. A continuum limit argument would require: (i) a fixed interface topology (number of pores and interfaces independent of h), (ii) shape-regular mesh refinement with stable trace spaces, and (iii) convergence of the discrete Steklov–Poincaré operator. The watershed-based meshes in the numerical study do* not *satisfy condition (i): the basin count depends on the mesh size. The observed h-insensitivity of the Affine-DDPNM errors within the tested range ($4.3K$–$45K$ tetrahedra) is therefore a numerical observation, not a consequence of a limiting argument. A rigorous h-convergence theory requires a fixed interface topology.*

Remark A.5 (Relation to monolithic FEM)**.** *The broken-pressure C-DD formulation used here differs from the monolithic continuous-P_1 Taylor–Hood FEM in its pressure space ($Q_h^{br} = \bigoplus_i Q_{i,h}$ vs. $Q_h^c \subset H^1$). For the geometries tested, the two references coincide numerically to $O(10^{-12})$ (verified via the exact FE-trace Schur oracle). The error estimate of Theorem A.2 is therefore a reliable proxy for the monolithic FEM error in practice, but a formal proof of equivalence between the two reference formulations is beyond the present scope.*

References

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